Teddy TRUONG

teddytruong@protonmail.com · teddytruong.dev · LinkedIn · GitHub

Software Engineer Intern

Professional Profile

Computer Science Master's student focused on systems programming and software engineering. Two years of industry experience shipping production C# tooling, Python pipelines, and automation at SNCF. Personal projects push into Rust, WebAssembly, parallel computing, and multi-agent simulation.

Technical Skills

Languages: Rust, Python, C#, C++, C, PHP, PowerShell, AutoLisp.

Databases: SQL, NoSQL.

Tools & Infrastructure: Git, GitHub Actions (CI/CD), Unity, Godot, Unreal Engine, Blender, Power BI, Figma.

Work Experience

UQAC – Unity Developer *(Fulltime)* December 2025 – Current

- Backend development of user management systems with NoSQL database integration.
- UI/UX redesign of core application interface.

SNCF Réseau – Software Engineer & Data Scientist *(Apprenticeship)* Oct. 2023 – Aug. 2025

- Diagnosed and eliminated flawed algorithms in a business-critical C# data pipeline. Redesigned core logic, rewrote tooling in Python with Power BI (Power Query + DAX), and built reporting dashboards — cutting execution time from 4+ hours to seconds.
- Developed CAD software add-ons (AutoCAD, BricsCAD) using C# (.NET) and VB.NET.
- Automated repetitive workflows using Power Automate and custom scripts.

Numérique Gagnant – Developer *(Apprenticeship)* Oct. 2022 – Aug. 2023

- Built automated archiving tools using PowerShell and Power Automate into SharePoint.
- Managed customer ticket resolution and hardware maintenance.
- Executed a major technical migration of a legacy website from PHP 3 to PHP 8.

Projects

MAFIS – Multi-Agent Fault Injection Simulator [GitHub] [Demo]

Fault resilience simulator for lifelong multi-agent pathfinding (MAPF). Built in Rust with Bevy/ECS, compiled to WebAssembly, runs entirely in the browser with deterministic replay and shareable URLs. Implements PIBT, RHCR, and Token Passing solvers with a differential metric framework to compare baseline vs. faulted runs. Covered by unit, integration, metamorphic, and calibration tests. Headless experiment runner parallelized with Rayon. CI/CD via GitHub Actions (WASM build + Vercel deploy on tag).

Education

UQAC (Canada) – M.Sc. in Computer Science (Double Degree) 2025 – 2026

ESGI (France) – Master in VR & Video Game Development 2021 – 2026

Lycée Dorian – CPGE PTSI (Intensive Prep School) 2020 – 2021

Awards & Competitions

Laval Virtual Student Competition – 3rd place out of 13 teams April 2025

Interests

Bouldering, FPS, Chess.